

Goal-Directed Knowledge Application and an Infrastructure for Realizing Desired Future States of Knowledge

MOTIVATION

Informatics has long been concerned with the application of knowledge to improve decision making and action. Recent advances in computable biomedical knowledge (CBK)¹⁻⁴, FAIR Digital Objects (FDOs)⁵, and artificial intelligence^{6,7} have increased the ability of systems to apply knowledge at scale. Yet the *goals* motivating knowledge application are often not represented formally within such systems. This paper proposes *goal-directed knowledge application* and a corresponding infrastructure for it, Goal-Directed Intelligence Layers (GDILs).

BUILDING ON KNOWLEDGBASES

The term *knowledgebase* emerged from artificial intelligence and expert systems research to distinguish collections of knowledge from collections of data.⁸ Whereas databases primarily organize observations, records, and transactions for retrieval, knowledgebases represent concepts, relationships, rules, models, and other logic for reasoning and inference. When coupled with runtimes, knowledgebases apply stored knowledge primarily to generate new inferences, but also recommendations, advice, or other relevant outputs. While newer information systems combine data and knowledge representations within graph structures⁹, the distinction remains useful. Databases help lookup facts. Knowledgebases help apply knowledge. GDILs align and apply knowledge in pursuit of desired future states of knowledge. GDILs differ from knowledgebases by using formalized goals to focus the application of inferential capability.

GOALS AND ARTIFICIAL INTELLIGENCE

Since the earliest decades of artificial intelligence, goals have been represented as desired future states toward which reasoning, planning, and action are directed.

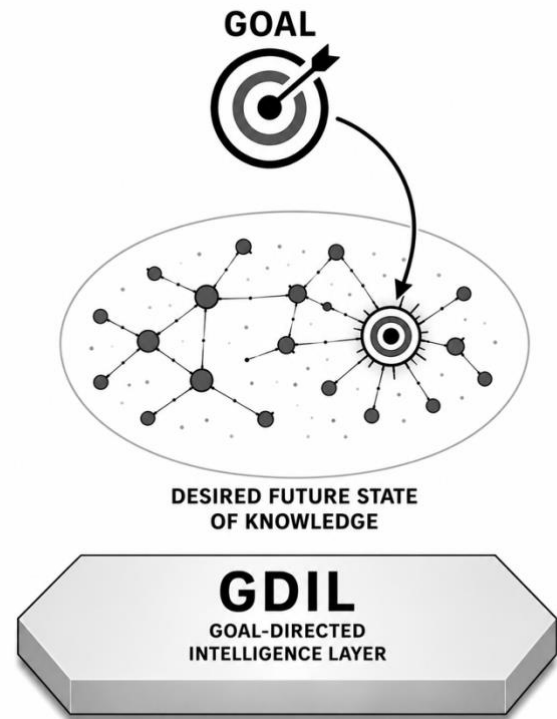


Figure 1. An explicit computational goal (top) specifies a desired future state of knowledge. An underlying Goal-directed Intelligence Layer includes a formalized representation of the goal and computable knowledge. It uses machine-processable metadata to align and apply computable knowledge in a goal-directed manner, achieving the goal and thus realizing the desired future state of knowledge, which is shown in the middle of the figure as a new addition to a knowledge graph.

From seminal work by McCarthy, Newell, Simon, and others^{10,11}, intelligent behavior was often understood in terms of selecting and applying methods capable of transforming a current state into a goal state. To attain this type of intelligence, GDILs begin by incorporating formalized computational goals. These goals provide a domain-independent abstraction for knowledge application that makes systematic alignment and application of knowledge possible across diverse use cases. The defining intelligence of GDILs is their ability to align and apply existing computable knowledge in a goal-directed way.

DESIRED FUTURE STATES OF KNOWLEDGE

Desired future states of knowledge are explicit representations of target knowledge states. Formalized computational goals specify desired future states of knowledge, and GDILs align and apply computable knowledge to achieve them. These key concepts are portrayed in Figure 1:

A formalized computational goal specifies the desired future state of knowledge.

The desired future state of knowledge is realized when the computational goal is achieved.

GDILs provide an underlying infrastructure for achieving computational goals.

EXPANDING THE ROLE OF METADATA

The idea of *goal-directed knowledge application* stems from prior investigations into CBK and its application. Our initial work examined the production, storage, and execution of CBK artifacts and related approaches for packaging them with metadata¹²⁻¹⁴. Subsequent work explored the composition and use of systematically packaged CBK as an interactive computational resource^{15,16}. Somewhat more recent efforts focused on the metadata required to describe, discover, and govern CBK artifacts, including their representation as FAIR Digital Objects¹⁷⁻¹⁹. This progression shifted attention from the computational capability of individual artifacts to how metadata facilitates their discovery, selection, and deployment while exposing provenance and execution history.

Insights from leveraging CBK metadata motivated the present conceptual work to make computational goals explicit, package them with metadata in Goal-bearing FAIR DOs alongside computable knowledge, and to put forward GDILs as infrastructure for achieving such formalized computational goals. GDILs expand the role of metadata from making computable knowledge findable, accessible, interoperable and reusable (FAIR²⁰) to making computable knowledge

suitable for selection, composition, and automatic application as a systematic means for achieving computational goals.

EXCLUSIVE FOCUS ON KNOWLEDGE APPLICATION

A distinction should be made between the generation of new knowledge and the application of existing knowledge. These are two distinct activities requiring different infrastructures, methods, and approaches to stewardship. Much contemporary work in artificial intelligence, machine learning, and knowledge discovery focuses on deriving new knowledge from data through the creation of models, predictions, or other inferential artifacts^{6,21}. In contrast, GDILs focus exclusively on applying existing computable knowledge to achieve explicit computational goals, not on generating new computable knowledge from collections of observations.

CORE ARGUMENT

The preceding discussion suggests a shift in perspective. Rather than focusing primarily on the representation, storage, or execution of computable knowledge, *goal-directed knowledge application*, enabled by GDILs, focuses on aligning knowledge application to achieve explicit computational goals. The following three propositions distill this core argument.

Proposition 1

Goal-directed knowledge application seeks to achieve desired future states of knowledge specified by formalized computational goals.

Proposition 2

Computable knowledge artifacts apply existing knowledge to generate new inferences that may assist in realizing desired future states of knowledge.

Proposition 3

Goal-Directed Intelligence Layers use metadata to align computable knowledge with goals and then apply it in pursuit of desired future states of knowledge.

ILLUSTRATIVE EXAMPLES OF KEY CONCEPTS

Table 1 lists goals and corresponding desired future states of knowledge.

Computational Goal	Desired Future State of Knowledge
<i>Determine whether a condition holds</i>	<i>Whether the condition holds is known</i>
<i>Estimate a quantity of interest</i>	<i>The estimated quantity is known</i>
<i>Classify an entity</i>	<i>The entity's classification is known</i>
<i>Rank two or more alternatives</i>	<i>The ranking of the alternatives is known</i>
<i>Select among alternatives</i>	<i>The selected alternative is known</i>

Table 1. Examples of Goal-DFSK Pairs

Table 2 lists metadata types. On the left are types for FAIRness. On the right are more types needed for *goal-directed knowledge application*.

Types of Metadata for Making Computable Knowledge FAIR	Additional Types of Metadata for Goal-Directed Knowledge Application
<i>Identity</i>	<i>Purpose</i>
<i>Discovery</i>	<i>Relevant Goals</i>
<i>Access</i>	<i>Relevant Desired Future States of Knowledge</i>
<i>Provenance</i>	<i>Knowledge Selection</i>
<i>Interoperability</i>	<i>Knowledge Composition</i>
<i>Reuse</i>	<i>Application Methods</i>
<i>Attribution</i>	<i>Knowledge Relationship</i>
<i>Rights and Governance</i>	<i>Stewardship</i>

Table 2. Metadata Types

Table 3 gives examples of goal-to-knowledge alignment. This alignment must precede computation to realize desired future states of knowledge using GDILs.

Computational Goal	Available Computable Knowledge	Alignment
<i>Determine value of X</i>	<i>Can produce value of X</i>	<i>Aligned</i>
<i>Determine value of X</i>	<i>Can produce value of Y</i>	<i>Not aligned</i>
<i>Classify X</i>	<i>Can produce classification of X</i>	<i>Aligned</i>
<i>Estimate X</i>	<i>Can first produce value of A and can then produce estimate of X based on value of A</i>	<i>Aligned</i>

Table 3. Examples of Goal–Knowledge Alignment

IMPLICATIONS FOR KNOWLEDGE SYSTEMS

The following implications start broadly and progressively narrow to show how *goal-directed knowledge application* relates to several operational contexts and scientific fields.

Learning Health Systems that apply knowledge to continuously improve health and healthcare provide a use case for GDILs²². In this view, GDILs complement rather than replace existing Learning Health System infrastructure by providing a platform with which formalized computational goals, computable knowledge, metadata, and FAIR Digital Objects may be brought together to support *goal-directed knowledge application*. As increasingly large bodies of computable knowledge become available, GDILs offer one possible approach for systematically aligning and applying that knowledge in support of continuous learning and improvement.

An infrastructural perspective helps explain why GDILs are conceptualized as “intelligence layers” rather than knowledgebases. Knowledgebases are typically designed to support single domains and their inferential tasks. GDILs, by contrast, are conceived as multi-purpose layers of knowledge infrastructure for operationalizing goal-directed knowledge application. Star and Ruhleder observed that infrastructure derives much of its value from supporting multiple communities and activities rather than any single use case²³. In this manner, GDILs offer extensible infrastructure that enables the same underlying computable knowledge to be reused across an open-ended range of downstream systems and purposes.

A practical implication concerns the economics of metadata generation. Metadata are costly to create and maintain, and have therefore relied heavily on public investment, institutional support, and community stewardship²⁴. Given their extensive metadata requirements, GDILs add to these costs. Within GDILs, metadata function as a core part of the computational infrastructure and thus constitute a foundational dependency. Advances in large language models and automated metadata generation may reduce

some metadata generation costs²⁵ but are unlikely to eliminate them. The long-term feasibility of GDILs will therefore depend upon sustainable approaches to metadata production, governance, and maintenance.

Along with new roles for metadata, GDILs introduce formalized computational goals specifying desired future states of knowledge as first-class objects of stewardship alongside computable knowledge itself. Foundational knowledge management research has illuminated what it takes to maintain knowledge assets, while knowledge management systems have provided support for doing so. Yet Ackerman's socio-technical gap²⁶ reminds us that no computational infrastructure can fully capture the contexts in which knowledge is ultimately interpreted and used to guide decisions and actions. GDILs require people to steward not only computable knowledge assets, but also the goals, metadata, and relationships through which computable knowledge is aligned and applied. They depend upon continuing human and institutional stewardship as computational goals, computable knowledge artifacts, and their real-world contexts of use evolve over time.

Computer science research has demonstrated the value of both predefined and dynamically composed approaches to computational orchestration.^{27,28} GDILs are compatible with both. Computable goal and knowledge artifacts, services, workflows, cases management artifacts, and orchestrations may be represented as FAIR Digital Objects and aligned with desired future states of knowledge regardless of whether their execution pathways are specified in advance, dynamically assembled at runtime, or formed through combinations of both approaches.

In a major finding from library science, Taylor observed that information seekers often do not directly express the goals underlying their information needs, requiring librarians to engage in a process of question negotiation to uncover the user's actual goal²⁹. This insight suggests that effective information seeking depends not only

on information resources, but also on gaining a good understanding of the goals motivating their use. The goal-directed knowledge application that GDILs support reflects Taylor's emphasis on stipulating the goals that motivate information seeking.

Finally, GDILs are anchored in at least one aspect of modern information philosophy. Floridi's Network Theory of Account³⁰ suggests that knowledge arises not from isolated informational objects, but from their participation in explanatory networks of relationships. GDILs adopt a similar emphasis on explicit relationships among Goal-bearing and CBK-bearing FAIR Digital Objects, using metadata to make relationships between these things machine processible. Therefore, GDILs can be viewed pragmatically as infrastructure in which graph-structured metadata make relationships among a mix of goal and knowledge artifacts explicit and machine processible, while setting aside broader epistemological questions concerning the relational nature of knowledge.

FUTURE WORK

A research plan to explore the feasibility of *goal-directed knowledge application* is under development. The plan is guided by a list of key research questions. The first of these relates to goal-knowledge alignment and asks, "To what degree and under which circumstances can metadata cause the right knowledge to be selected for the right goal?" An initial technical demonstration is now being pursued to answer both parts of this question. Additional future work to establish a role for GDILs in realizing desired future states of knowledge awaits these initial results.

CONCLUSION

By returning to the concept of goals that was once central to artificial intelligence, and focusing specifically on formalized computational goals, it becomes possible to build on the concept of knowledgebases in an infrastructural direction to

arrive at Goal-Directed Intelligence Layers (GDILs). GDILs are infrastructure for goal-directed knowledge application, which relies on metadata to align and apply computable knowledge to realize desired future states of knowledge. Unlike approaches concerned with generating new knowledge from many observations, GDILs assume the existence of computable knowledge. The implications of *goal-directed knowledge application* may include new infrastructure capable of applying large bodies of computable knowledge at scale across an open-ended range of domains and downstream applications.

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